



HikCameraSDK_AUO

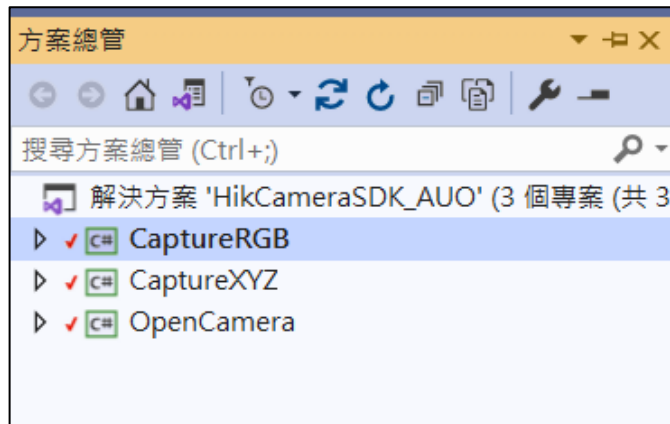
2026/5



SDK Sample Code

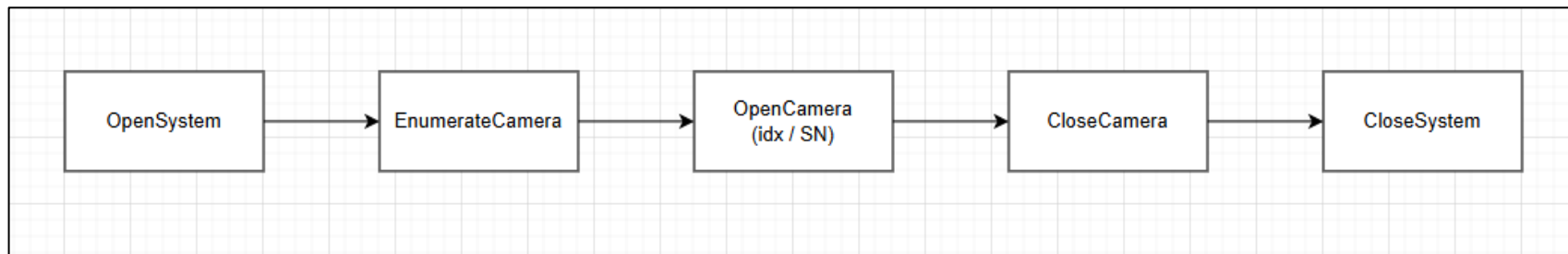
- 包含3組Sample Code

1. OpenCamera: 列舉並開啟目前連結到的相機
2. Capture RGB: 取得RGB影像
3. Capture XYZ: 取得XYZ影像



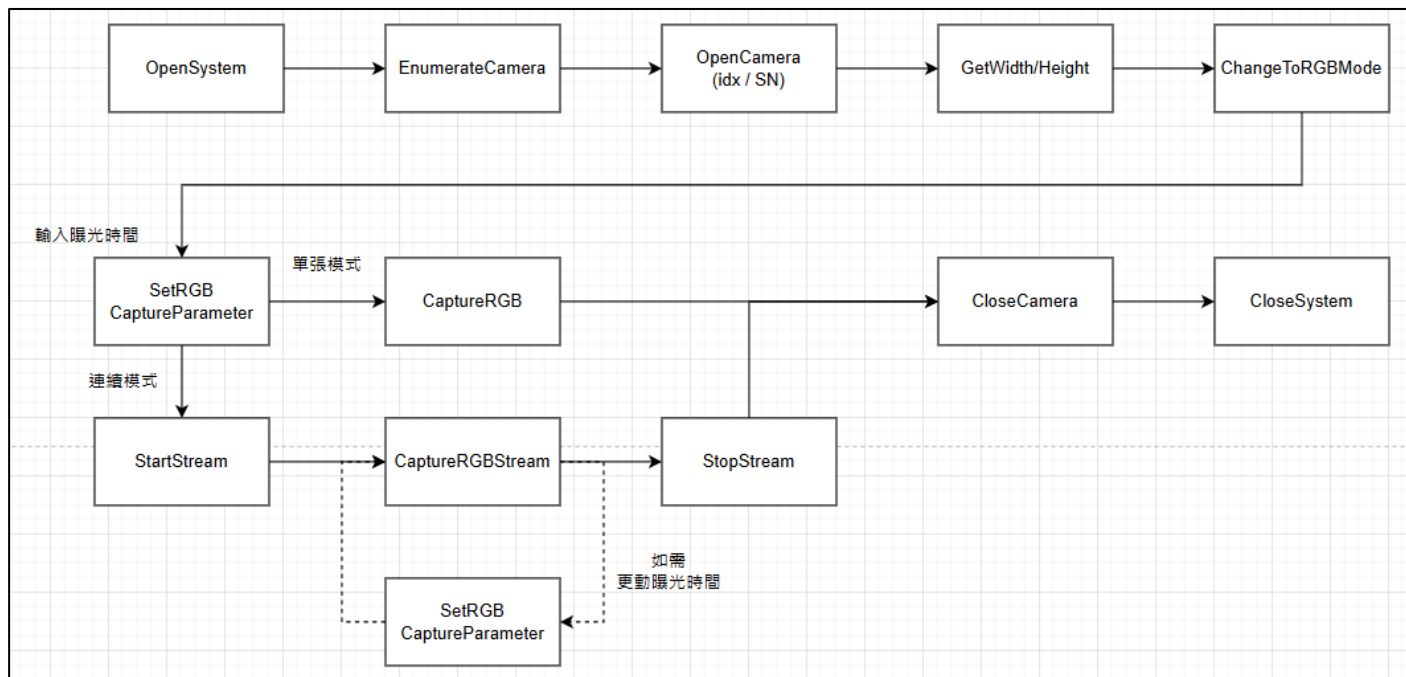
OpenCamera

- API呼叫順序



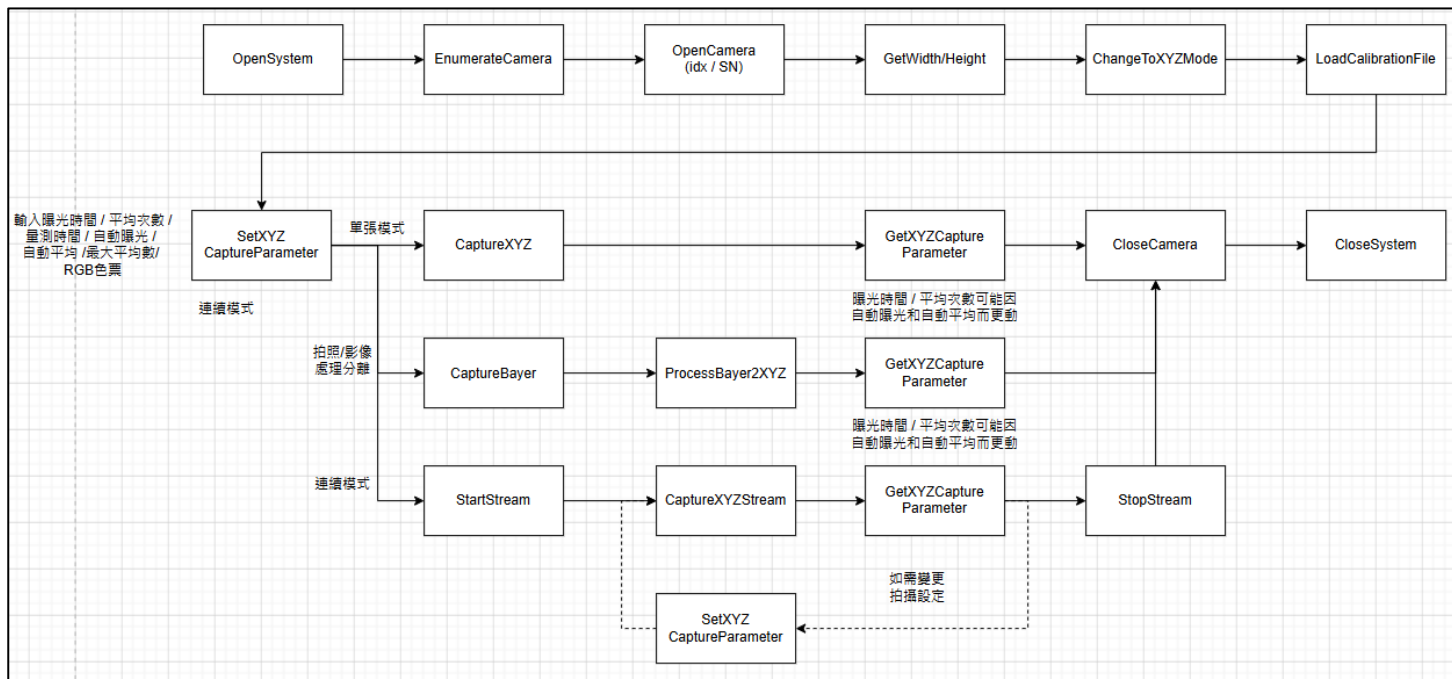
CaptureRGB/Stream

● API呼叫順序



CaptureXYZ/Stream/拍照處理分離

● API呼叫順序



Function API

- **OpenSystem:** 初始化相機系統
 - `ERRORCODE OpenSystem (camhandle_t* camhandle);`
- **CloseSystem:** 關閉相機系統
 - `ERRORCODE CloseSystem (camhandle_t* camhandle);`
- **EnumerateCamera:** 列舉並讀取相機資訊
 - `ERRORCODE EnumerateCamera (camhandle_t camhandle, CameraInfo* infoArray, int* count);`
- **OpenCamera:** 開啟指定idx/SN的相機
 - `ERRORCODE OpenCameraIdx (camhandle_t camhandle, int cam_idx);`
 - `ERRORCODE OpenCameraSN (camhandle_t camhandle, const char* cam_sn);`
- **CloseCamera:** 關閉相機
 - `ERRORCODE CloseCamera (camhandle_t camhandle);`
- **GetWidth & GetHeight:** 取得相機的長寬
 - `ERRORCODE GetWidth (camhandle_t camhandle, int* width);`
 - `ERRORCODE GetHeight (camhandle_t camhandle, int* height);`
- **LoadCalibrationFile:** 讀取相機的校正檔
 - `ERRORCODE LoadCalibrationFile (camhandle_t camhandle, const char* file_path);`



Function API

- **StartStream & StopStream** 開啟/關閉即時串流

- `ERRORCODE StartStream (camhandle_t camhandle);`
- `ERRORCODE StopStream (camhandle_t camhandle);`

- **ChangeToXYZMode & ChangeToRGBMode** : 更改相機模式

- `ERRORCODE ChangeToXYZMode (camhandle_t camhandle);`
- `ERRORCODE ChangeToRGBMode (camhandle_t camhandle);`

- **變更RGB, XYZ拍照設定**

- `ERRORCODE SetRGBCaptureParameter(camhandle_t camhandle, double exp_us);`
- `ERRORCODE SetXYZCaptureParameter(camhandle_t camhandle, double exp_us, int avg, double measure_time, bool auto_exp, bool auto_avg, int max_avg, int pattern_R, int pattern_G, int pattern_B);`
 - ◆ 手動模式
 - 輸入曝光時間與平均次數 (`exp_us`, `avg`)
 - ◆ 自動模式
 - 自動平均 (`auto_avg = true`) · 平均次數最多`max_avg`次 · 需要設定`measure_time_us`
 - 自動曝光 (`auto_exp = true`)

- **取得XYZ拍攝實際的曝光時間/平均次數**

- `ERRORCODE GetXYZCaptureParameter(camhandle_t camhandle, double* exp_us, int* avg);`



Function API

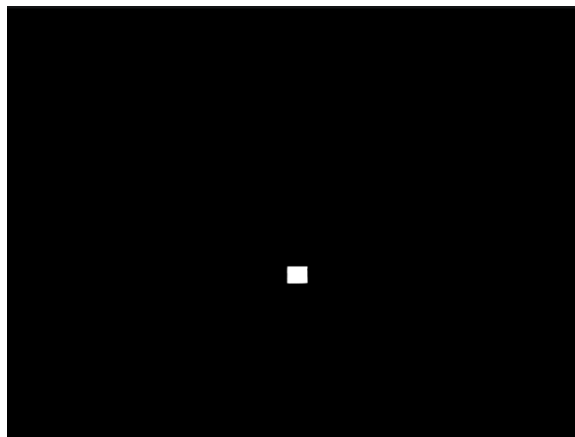
- **CaptureRGB: 取得相機的單張/即時影像 (RGB uint8 3 channel)**
 - 單張: `ERRORCODE CaptureRGB (camhandle_t camhandle, uint8_t* img_buffer);`
 - 連續: `ERRORCODE CaptureRGBStream (camhandle_t camhandle, uint8_t* img_buffer);`
- **CaptureXYZ: 取得XYZ影像/即時影像 (XYZ float32 3channel)**
 - 單張: `ERRORCODE CaptureXYZ (camhandle_t camhandle, float* img_buffer);`
 - 連續: `ERRORCODE CaptureXYZStream (camhandle_t camhandle, float* img_buffer);`
- **拍攝Bayer影像**
 - `ERRORCODE CaptureBayer(camhandle_t camhandle, float* img_buffer);`
- **處理Bayer影像到XYZ**
 - `ERRORCODE ProcessBayer2XYZ(camhandle_t camhandle, float* img_buffer_bayer, float* img_buffer_XYZ, double exp_us, int pattern_R, int pattern_G, int pattern_B);`
- **取得校正ROI區域數量與座標**
 - `ERRORCODE GetROIs(camhandle_t camhandle, ROI* rois);`
 - `ERRORCODE GetROIsNum(camhandle_t camhandle, int* roi_num);`



校正檔產生

● CameraPoints

- 利用產生的校正Excel，來針對ROI進行校正
- 未來SDK輸出的XYZ影像只有會ROI的數值，其餘為0
- 可以利用多筆Excel來產生校正檔h5



XYZ影像

1	F-number	4.5
2	Focus	0
3	Image Wi	14208
4	Image He	10640
5	Camera SN	
6	roi1_x	6906
7	roi1_y	6376
8	roi2_x	7468
9	roi2_y	6908
10		
11		
12		
13		
14		
15		
16		
17		

校正Excel (MetaData下)

校正檔產生

校正檔資訊

F# 1.8

Focus 0

Apply

先讀入鏡頭校正檔

校正點位

Threshold ☐ Auto 1.00

膨脹/侵蝕 50 50

Mask 建立 檢查 讀取

點位 1

Color No Color << >>

MetaV 0,0,0 (100us)

Standard 0,0,0

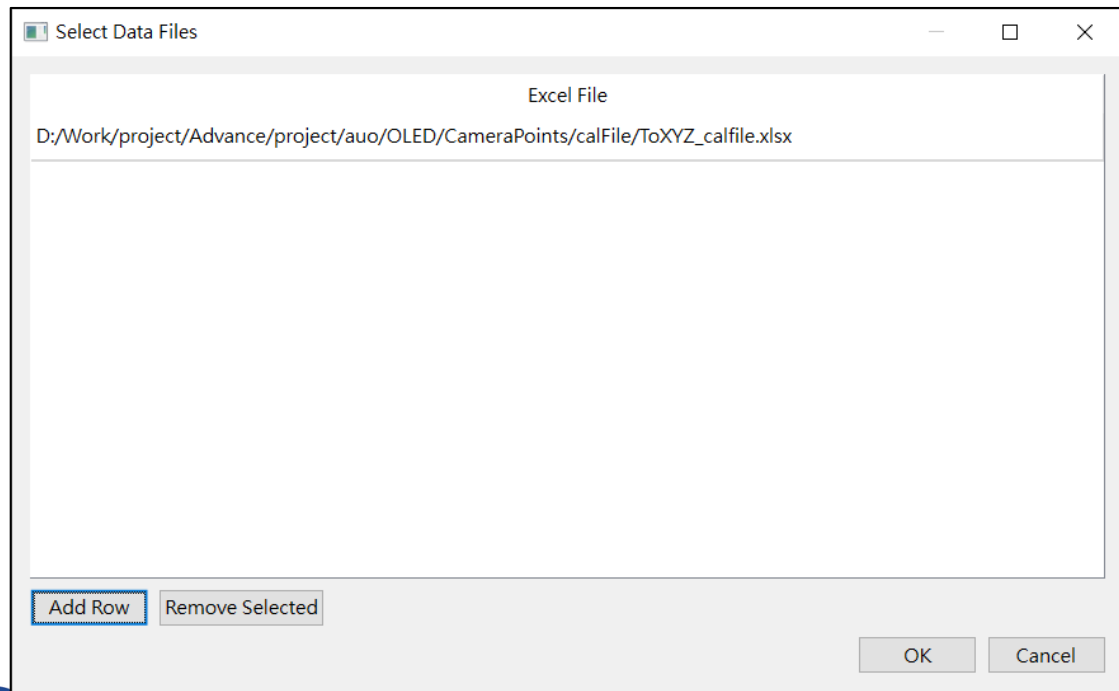
Start Stop Save Load ☐ CC

CA Robot **OutputCal**

色彩校正

按下OutputCal開始建立校正檔

校正檔產生



利用Add Row讀入校正Excel

1	F-number	4.5
2	Focus	0
3	Image Wi	14208
4	Image He	10640
5	Camera SN	
6	roi1_x	6906
7	roi1_y	6376
8	roi2_x	7468
9	roi2_y	6908
10		
11		
12		
13		
14		
15		
16		
17		

MetaData Point_1

確保校正ROI已經設定

校正檔產生

- 在Out對應的資料夾下，會產生H5檔案，可以供SDK使用

使用者 > cal > 桌面 > CameraPoints > out > 20260514_192142 >				搜尋 20260514_19214
名稱	修改日期	類型	大小	
XYZ_img	2026/5/14 下午 07:21	檔案資料夾		
XYZ_raw	2026/5/14 下午 07:21	檔案資料夾		
calibration_data_20260514_192234.h5	2026/5/14 下午 07:22	H5 檔案	1,919,271 KB	

OpenCamera (By IDX)

```
Microsoft Visual Studio Debug Console
=== Demo 1: Open Camera ===

Step 1: Opening system...
System opened successfully.

Step 2: Enumerating cameras...
Found 1 camera(s):
  Camera 0: SN=DA7468833

Choose how to open the camera:
  1. By Index (e.g., 0, 1, 2...)
  2. By Serial Number
Enter your choice (1 or 2): 1
Enter camera index (default 0): 0 ←
Step 3: Opening camera at index 0...
Camera opened successfully.

Press any key to close camera...

Step 4: Closing camera...
Camera Close
Camera closed successfully.

Step 5: Closing system...
System closed successfully.

Press any key to exit...

C:\Users\cal\Downloads\HikCameraSDK_AUO (1)\OpenCamera\bin\Debug\OpenCamera.exe (process 32696) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugg
ing stops.
Press any key to close this window . . .
```

OpenCamera (By SN)

```
Microsoft Visual Studio Debug Console
=== Demo 1: Open Camera ===

Step 1: Opening system...
System opened successfully.

Step 2: Enumerating cameras...
Found 1 camera(s):
  Camera 0: SN=DA7468833

Choose how to open the camera:
  1. By Index (e.g., 0, 1, 2...)
  2. By Serial Number
Enter your choice (1 or 2): 2
Enter camera Serial Number: DA7468833
Step 3: Opening camera with SN: DA7468833...
Camera opened successfully.

Press any key to close camera...

Step 4: Closing camera...
Camera Close
Camera closed successfully.

Step 5: Closing system...
System closed successfully.

Press any key to exit...

C:\Users\cal\Downloads\HikCameraSDK_AUO (1)\OpenCamera\bin\Debug\OpenCamera.exe (process 7952) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

CaptureRGB

```
Microsoft Visual Studio Debug Console
=== Demo 2: Capture RGB / Stream ===

Step 1: Opening system...
System opened successfully.

Step 2: Enumerating cameras...
Found 1 camera(s):
  Camera 0: SN=DA7468833

Choose how to open the camera:
  1. By Index (e.g., 0, 1, 2...)
  2. By Serial Number
Enter your choice (1 or 2): 1
Enter camera index (default 0): 0
Step 3: Opening camera at index 0...
Camera opened successfully.

Step 4: Getting image dimensions...
Image size: 14208 x 10640

Step 5: Changing to RGB mode...
Changed to RGB mode successfully.

Select capture mode:
  1. Single Capture (CaptureRGB)
  2. Stream Capture (CaptureRGBStream)
Enter choice (1 or 2): 1
Enter exposure time in microseconds (default 10000): 3000000

Step 6: Setting RGB capture parameters...
Parameters set - Exposure: 3000000 us

Step 7: Capturing single RGB image... ←
RGB image captured successfully.

Image saved to: captured_rgb.bmp
Press any key to close camera...

Closing camera...
```



CaptureRGBStream

```
Microsoft Visual Studio Debug Console
System opened successfully.

Step 2: Enumerating cameras...
Found 1 camera(s):
  Camera 0: SN=DA7468833

Choose how to open the camera:
  1. By Index (e.g., 0, 1, 2...)
  2. By Serial Number
Enter your choice (1 or 2): 1
Enter camera index (default 0): 0
Step 3: Opening camera at index 0...
Camera opened successfully.

Step 4: Getting image dimensions...
Image size: 14208 x 10640

Step 5: Changing to RGB mode...
Changed to RGB mode successfully.

Select capture mode:
  1. Single Capture (CaptureRGB)
  2. Stream Capture (CaptureRGBStream)
Enter choice (1 or 2): 2
Enter exposure time in microseconds (default 10000): 3000000

Step 6: Setting RGB capture parameters...
Parameters set - Exposure: 3000000 us

Step 7: Starting stream...
Stream started successfully.

Enter number of frames to capture (default 10): 2

Step 8: Capturing 2 RGB stream frames...
Frame 1/2 captured (exposure: 3000000 us)
Frame 2/2 captured (exposure: 3001000 us)
Image saved to: captured_rgb_stream_frame_2.bmp
Stream capture completed.
```



CaptureXYZ (W)

```
D:\CameraPoints\測試\csharp_sample\CaptureXYZ\bin\Debug\CaptureXYZ.exe
Enter exposure time in microseconds (default 10000): 1243487
Enter averaging count (default 1): 5
Enter measurement time in microseconds (default 1000000): 5000000
Enable auto exposure? (y/n, default n): n
Enable auto averaging? (y/n, default n): n
Enter max average number (default 5): 5
Enter Red pattern value (default 255): 255
Enter Green pattern value (default 255): 255
Enter Blue pattern value (default 255): 255

Select capture mode:
1. Single Capture (CaptureXYZ)
2. Stream Capture (CaptureXYZStream)
3. Bayer Capture (CaptureBayer + ProcessBayer2XYZ)
Enter choice (1, 2, or 3): 3

Step 6: Setting XYZ capture parameters...
Parameters set - Exposure: 1243487 us, Averaging: 5

Step 7: Capturing Bayer image (stream managed internally)...
CaptureBayer processing completed in 6949 ms.

Bayer image captured successfully.
Actual parameters - Exposure: 1243487 us, Averaging: 5

Step 8: Processing Bayer to XYZ...
ProcessBayer2XYZ processing completed in 1083 ms.

XYZ data saved to: captured_xyz_bayer.tiff
- File size: 1814077920 bytes
- Format: 32-bit float ImageJ TIFF
- Dimensions: 14208x10640x3

--- ROI XYZ Averages (2 ROIs) ---
ROI[0] (6906,6376)-(7468,6908): X=60.7747, Y=63.6799, Z=68.4133, L=63.6799, x=0.31511, y=0.33017
ROI[1] (5036,6315)-(5599,6878): X=0.0847, Y=0.0889, Z=0.0958, L=0.0889, x=0.31425, y=0.33001

Press any key to close camera...
```

CaptureXYZ (R)

```
D:\CameraPoints\測試\csharp_sample\CaptureXYZ\bin\Debug\CaptureXYZ.exe
Enter exposure time in microseconds (default 10000): 2667023
Enter averaging count (default 1): 5
Enter measurement time in microseconds (default 1000000): 5000000
Enable auto exposure? (y/n, default n): n
Enable auto averaging? (y/n, default n): n
Enter max average number (default 5): 5
Enter Red pattern value (default 255): 255
Enter Green pattern value (default 255): 0
Enter Blue pattern value (default 255): 0

Select capture mode:
1. Single Capture (CaptureXYZ)
2. Stream Capture (CaptureXYZStream)
3. Bayer Capture (CaptureBayer + ProcessBayer2XYZ)
Enter choice (1, 2, or 3): 3

Step 6: Setting XYZ capture parameters...
Parameters set - Exposure: 2667023 us, Averaging: 5

Step 7: Capturing Bayer image (stream managed internally)...
CaptureBayer processing completed in 14076 ms.

Bayer image captured successfully.
Actual parameters - Exposure: 2667023 us, Averaging: 5

Step 8: Processing Bayer to XYZ...
ProcessBayer2XYZ processing completed in 1081 ms.

XYZ data saved to: captured_xyz_bayer.tiff
- File size: 1814077920 bytes
- Format: 32-bit float ImageJ TIFF
- Dimensions: 14208x10640x3

--- ROI XYZ Averages (2 ROIs) ---
ROI[0] (6906,6376)-(7468,6908): X=25.8608, Y=13.6510, Z=1.0269, L=13.6510, x=0.63793, y=0.33674
ROI[1] (5036,6315)-(5599,6878): X=0.0376, Y=0.0200, Z=0.0010, L=0.0200, x=0.64218, y=0.34059

Press any key to close camera...
```

CaptureXYZ (G)

```
D:\CameraPoints\測試\csharp_sample\CaptureXYZ\bin\Debug\CaptureXYZ.exe
Enter exposure time in microseconds (default 10000): 1765941
Enter averaging count (default 1): 5
Enter measurement time in microseconds (default 1000000): 5000000
Enable auto exposure? (y/n, default n): n
Enable auto averaging? (y/n, default n): n
Enter max average number (default 5): 5
Enter Red pattern value (default 255): 0
Enter Green pattern value (default 255): 255
Enter Blue pattern value (default 255): 0

Select capture mode:
1. Single Capture (CaptureXYZ)
2. Stream Capture (CaptureXYZStream)
3. Bayer Capture (CaptureBayer + ProcessBayer2XYZ)
Enter choice (1, 2, or 3): 3

Step 6: Setting XYZ capture parameters...
Parameters set - Exposure: 1765941 us, Averaging: 5

Step 7: Capturing Bayer image (stream managed internally)...
CaptureBayer processing completed in 9566 ms.

Bayer image captured successfully.
Actual parameters - Exposure: 1765941 us, Averaging: 5

Step 8: Processing Bayer to XYZ...
ProcessBayer2XYZ processing completed in 1086 ms.

XYZ data saved to: captured_xyz_bayer.tiff
- File size: 1814077920 bytes
- Format: 32-bit float ImageJ TIFF
- Dimensions: 14208x10640x3

--- ROI XYZ Averages (2 ROIs) ---
ROI[0] (6906,6376)-(7468,6908): X=23.5516, Y=46.4898, Z=4.6244, L=46.4898, x=0.31543, y=0.62264
ROI[1] (5036,6315)-(5599,6878): X=0.0514, Y=0.0979, Z=0.0101, L=0.0979, x=0.32263, y=0.61430

Press any key to close camera...
```

CaptureXYZ (B)

```
D:\CameraPoints\測試\csharp_sample\CaptureXYZ\bin\Debug\CaptureXYZ.exe
Enter exposure time in microseconds (default 10000): 3085222
Enter averaging count (default 1): 5
Enter measurement time in microseconds (default 1000000): 5000000
Enable auto exposure? (y/n, default n): n
Enable auto averaging? (y/n, default n): n
Enter max average number (default 5): 5
Enter Red pattern value (default 255): 0
Enter Green pattern value (default 255): 0
Enter Blue pattern value (default 255): 255

Select capture mode:
1. Single Capture (CaptureXYZ)
2. Stream Capture (CaptureXYZStream)
3. Bayer Capture (CaptureBayer + ProcessBayer2XYZ)
Enter choice (1, 2, or 3): 3

Step 6: Setting XYZ capture parameters...
Parameters set - Exposure: 3085222 us, Averaging: 5

Step 7: Capturing Bayer image (stream managed internally)...
CaptureBayer processing completed in 16173 ms.

Bayer image captured successfully.
Actual parameters - Exposure: 3085222 us, Averaging: 5

Step 8: Processing Bayer to XYZ...
ProcessBayer2XYZ processing completed in 1081 ms.

XYZ data saved to: captured_xyz_bayer.tiff
- File size: 1814077920 bytes
- Format: 32-bit float ImageJ TIFF
- Dimensions: 14208x10640x3

--- ROI XYZ Averages (2 ROIs) ---
ROI[0] (6906,6376)-(7468,6908): X=12.9493, Y=5.1208, Z=64.5228, L=5.1208, x=0.15678, y=0.06200
ROI[1] (5036,6315)-(5599,6878): X=0.0223, Y=0.0089, Z=0.1120, L=0.0089, x=0.15572, y=0.06213

Press any key to close camera...
```

與 CS-200 驗證誤差

R	G	B	Exp(us)	avg	Capture(ms)	Process(ms)	Cam X	Cam Y	Cam Z	STD X	STD Y	STD Z	L(%)	dx	dy
255	255	255	1243487	5	6949	1083	60.7747	63.6799	68.4133	61.02	64.02	68.45	-0.53%	-0.0003	-0.0007
255	0	0	2667023	5	14076	1081	25.8608	13.6510	1.0269	25.86	13.61	0.98	0.30%	-0.0014	0.0003
0	255	0	1765941	5	9566	1086	23.5516	46.4898	4.6244	23.63	46.71	4.55	-0.47%	-0.0001	-0.0011
0	0	255	3085222	5	16173	1081	12.9493	5.1208	64.5228	12.99	5.12	64.68	0.02%	-0.0001	0.0002



共同的行動準則

— ∞ *Purposive • Conclusive • Productive* ∞ —

不斷的追求卓越